

Regular language

 $\cup$

or

.

*

Zero or more

$R_1 \cup R_2$ = The Union of 2 language, is the language generated by this RE

$R_1 . R_2$ = The Concatenation of 2 language, is the language generated by this RE

$(R)^*$ = The language kleene star, generated by this RE.

The language which this regular expression generates

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The kleene Closure

If $L = \{0, 1\}$

$L^* = \{L^0, L^1, L^2, L^3, \dots\}$

→ This continues till infinity.

Here,

$L^0 = \epsilon$

$L^1 = \{0, 1\}$

$L^2 = \{0, 1\} \{0, 1\} = \{00, 01, 10, 11\}$

$L^3 = \{0, 1\} \{0, 1\} \{0, 1\} = \{000, 001, 010, 011, 100, 101, 110, 111\}$

$$\textcircled{1} L = \{ w \in \{0,1\}^* : w \text{ starts with } 1 \}$$

$$1011 \in L$$

$$0000 \notin L$$

$$1(0 \cup 1)^* \rightarrow 0 \text{ or } 1, \text{ however many.}$$

$$\textcircled{2} L = \{ w \in \{0,1\}^* : w \text{ contains } 111 \}$$

$$(0 \cup 1)^* 111 (0 \cup 1)^*$$

Can match any binary string

$$\textcircled{3} L = \{ w \in \{0,1\}^* : w \text{ does not end with } 11 \}$$

$$(0 \cup 1)^* (00 \cup 01 \cup 10)$$

$$0 \in L \text{ [Not generated by this RE]}$$

$$1 \in L \quad //$$

$$\epsilon \in L \quad //$$

$$\epsilon \cup 0 \cup 1 \cup (0 \cup 1)^* (00 \cup 01 \cup 10)$$

④ $L = \{w \in \{0,1\}^* : \text{the length of } w \text{ is even}\}$

$(00^*01^*10^*11^*)^*$

If the length of a string is even, then it has to be the concatenation of zero or more length 2 strings.

Odd?

000010


Odd + odd = Even.

Odd + even = Odd

Odd: $(00^*01^*10^*11^*)^* (0^*1)$

⑤ $L = \{w \in \{0,1\}^* : \text{the length of } w \text{ can be expressed as } 3k+2 \text{ where } k \text{ is an integer}\}$

2, 5, 8, 11.

$\underbrace{((0^*1)(0^*1)(0^*1))^*}_{00110101} (0^*1)(0^*1)$

00110101

$$\textcircled{6} \quad L = \{ w \in \{0,1\}^* : w \text{ does not contain } 11 \}$$

00010100011 $\sim$ missing 1 in RE

Everytime I see a 0, 1 has to follow

$$(0 \cup 10)^* (\epsilon \cup 1)$$

$\hookrightarrow 1$ optional

$$\textcircled{7} \quad L = \{ w \in \{0,1\}^* : w \text{ does not contain } 00 \}$$

$$(1 \cup 01)^* (\epsilon \cup 0)$$

$$\textcircled{8} \quad L = \{ w \in \{0,1\}^* : w \text{ does not contain } 111 \}$$

$$(0 \cup 10 \cup 110)^* (\epsilon \cup 1 \cup 11)$$

Bracu email design

first. middle. last @ bracu.ac.bd.

Let $\Sigma = \{a, ., @\}$, a represent some letter.

$$aa^* (.aa^*)^* @ aa^* . aa^* . aa^*$$

$$a^+ (.a^+)^* @ a^+ . a^+ . a^+$$

$$a^+ (.a^+)^* @ (gve) . a^+ . a^+ . a^+$$

$$a^+ (.a^+)^* @ g? a^+ . a^+ . a^+$$